

Decision Memo — Onion Selling Windows, Maharashtra

To: Advisory Team, DeHaat

From: Ritik Date: 25 July 2026 Re: When, where, and which onion farmers should sell this season

The question

Which selling window, which market, and which onion variety should field agents recommend to FPO members selling onions in Maharashtra — and what will the price do next month?

The headline finding

Timing is worth about twice what market choice is worth, and variety is worth almost as much as market.

The same lot sold in **November** rather than **May** fetches **■1,503/quintal more** — a 2.5× swing. Choosing the best market over the worst is worth **■756/quintal** (gross of transport). Growing **Red** onion rather than **Local** is worth about **■530/quintal**. A farmer who optimises location while selling in the wrong month leaves most of the money on the table.

All 14 curated markets peak in **September–November** and trough in **April–May**; the seasonal pattern is common across the state, and only the price level differs.

Selling windows the data supports

- 1. Sell September–November. Prioritise November; July is a secondary window.** Curated-market average: **■2,495/quintal in November** against **■992 in May**. Every market ranks the Sep–Nov window first. *Confidence: high on direction, moderate on magnitude — 24 months only.*
- 2. Prefer the Nashik belt (Lasalgaon, Pimpalgaon) — if transport is cheap.** Lasalgaon markets average **■1,960–2,030/quintal (+12–16% vs state)**; Chhatrapati Sambhajinagar sits at **■1,273 (–27%)**. The **■756/quintal** spread is **gross** of transport, which is not in this data and must be netted off before advising a farmer to travel. *Confidence: high. Price rankings are stable across the record.*
- 3. Where feasible, grow and market Red onion.** Red averages **■2,231/quintal**, against Local at **■1,701** and Unhali at **■1,328** — a **~31% premium** that holds across months. *Confidence: high on ranking; a crop-planning lever, not a sell-timing one.*
- 4. Next month, expect prices near this month's, market by market.** The forecaster (walk-forward MAE **■168/quintal, 13.9%**) ranks **Pimpalgaon Baswant** highest for **July 2026** at **■1,788/quintal** (range **■1,555–1,884**). The dashboard carries the full ranking with an uncertainty band on each market.

Recommendation

Field agents should advise members to **plan around a September–November sale**, treating July as an earlier acceptable window, and to **avoid April–May sales where storage allows** — the critical qualifier. The recommendation is only actionable for farmers who can physically hold stock; for those who cannot, market and variety choice are the levers left, and they are worth less than timing.

Before this reaches farmers, confirm three things: that cold-storage capacity is actually available to the members being advised; that transport cost to the Nashik belt is lower than the **■756/quintal premium**; and that the variety advice fits the member's agronomy.

Uncertainty, stated plainly

Next month's price is close to unpredictable beyond "near this month's level." Monthly onion prices behave like a random walk with large, unforecastable shocks: a purpose-built model barely improves on assuming next month equals this month (**■168 vs ■169 mean error**). The value of the model is the **uncertainty band** it puts around each forecast, not a false claim of precision.

No model here can see a supply shock. The dataset records that prices spiked to **■25,000/quintal** in 2024–25 but contains nothing about *why* — no weather, no MSP, no export policy. The error runs one way: in a shock year the real peak will be **higher** than any forecast here, never lower. Advice built on it is conservative, not optimistic.

Limitations

The seasonal pattern rests on 24 months — each month observed ~2 times, and August 2024 is missing entirely. The November peak is estimated from few observations. This data cannot separate "November is reliably the peak" from "two unusual Novembers." This is the single largest caveat on the recommendation.

What would change my conclusion: five to seven years of history. **What I would want next**, in priority order: (1) district rainfall from IMD, (2) sowing acreage by season, (3) dated MSP and export-policy announcements, (4) cold-storage capacity by district, (5) farm-to-mandi transport costs. Items 1–3 address the shock blindness; items 4–5 determine whether the advice is actionable for a given farmer.

*Analysis: 31,930 daily price records, 111 Maharashtra mandis (14 curated),

Jun 2024 – Jun 2026, Agmarknet. Seasonal and market effects from OLS with robust standard errors; next-month forecast from a hybrid persistence + LightGBM model with quantile prediction intervals, evaluated walk-forward. Full methodology, data-quality note, and model card in the repository.*